

Twelve-week biomechanical ankle platform system training on postural stability and ankle proprioception in subjects with unilateral functional ankle instability

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Received 5 August 2007; accepted 23 April 2008

Abstract

Background. Functional ankle instability is defined as the subjective sensation of giving way or feeling joint instability after repeated episodes of ankle sprain. The purpose of this study was to examine the effects of 12-week biomechanical ankle platform system training on static postural stability and ankle reposition sense in subjects with unilateral functional ankle instability.

Methods. Twelve university students (4 females and 8 males) with unilateral functional ankle instability volunteered as subjects. The active and passive reposition senses were assessed using an isokinetic dynamometer. The mean radius of the center of pressure excursion was recorded during single-leg standing with a force platform. A 12-week training program and a progression test for controlling the platform in certain directions and advancing to next training level was given to each subject. Repeated-measures 2-way analyses of variance were conducted to determine differences in postural stability and ankle proprioception between each limb before and after the training period.

Findings. The mean radius of center of pressure on unilateral standing and the absolute error from pre-selected ankle angle in the functional ankle instability limb were significantly reduced after 12 weeks of training.

Interpretation. These improvements in postural stability appear to reflect improved neuromuscular ability along with enhanced functional joint stability, as ankle proprioception also demonstrated the same positive improvements after training.

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Keywords: Ankle instability; Sensorimotor; Balance control; Reposition sense

1. Introduction

Ankle sprains are the most common injuries in running and jumping sports, such as basketball, soccer, badminton, and volleyball (Payne et al., 1997). A previous study found that after an ankle sprain, up to 40% of sufferers continue to report residual disability (Safran et al., 1999) which might persistent for seven years after the inversion trauma (Konradsen et al., 2002). The term “Functional ankle instability” (FAI) was firstly purposed and coined by Free-

mann (1965) to describe the subjective sensation of giving way or feeling joint instability after repeated ankle sprain episodes. The factors contributing to FAI are complex but involve sensorimotor, mechanical, and muscular deficiencies (Hertel, 2000; Kaminski et al., 2003; Konradsen and Magnusson, 2000).

Postural stability has been defined as the ability to maintain an upright posture and to keep the center of gravity (COG) within the limits of the base of support (Karlsson and Frykberg, 2000). It is preserved through the dynamic integration of internal and external forces and factors involving the environment (Jonsson et al., 2004; Mochizuki et al., 2006). Postural stability has traditionally been exam-

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ined by spatial measures of the center of pressure (COP), where larger values of COP distributions are considered signs of postural instability (Lee and Lin, 2007). As a lower extremity joint close to the body's base of support, the ankle plays an integral role in postural control and balance maintenance (Blackburn et al., 2000).

Proprioception is a specialized variation of the sensory modality and encompasses the sensations of joint movement (kinesthesia) and joint position (joint reposition sense). It contributes to motor programming for neuromuscular control and contributes to muscle reflexes for dynamic joint stability (Lephart et al., 1997). It is believed that ankle proprioception is critical to the balance of the human body during functional activities such as standing, walking and running (Leanderson et al., 1996; Robbins and Waked, 1998). It has been demonstrated that subjects with FAI show impaired postural stability, decreased proprioception and altered peripheral control of ankle muscle function when compared with non-impaired subjects (Amaral De Noronha and Borges, 2004; Hertel, 2002; Willems et al., 2002).

Damage to the proprioception system is thought to be the major causes of functional instability after ankle injuries. The biomechanical ankle platform system (BAPS) has been commonly used by athletic trainers and physicians in rehabilitation after ankle injuries. It is designed to aid in re-educating the proprioceptive system by improving mechanoreceptor function and restoring normal neuromuscular coordination (Amaral De Noronha and Borges, 2004; Clark and Burden, 2005; Hoffman and Payne, 1995; Rozzi et al., 1999). The effectiveness of 4–8 weeks of wobble board training on postural control and perceived stability has been well documented (Rozzi et al., 1999; Bernier and Perrin, 1998; Hertel, 2000; Clark and Burden, 2005), and residual symptoms following ankle sprains can be reduced by a 12-week wobble board training program (Wester et al., 1996).

One study demonstrated that a 10-week BAPS training program significantly improved postural control (Hoffman and Payne, 1995). However, to our knowledge, no study has examined the training effects of BAPS on ankle proprioception. Furthermore, there is still some controversy as to the length of the training program (4, 6, 8, and 12 weeks) required for adaptations to occur (Riemann et al., 2003). Therefore, the purpose of this study was to examine the effects of a 12-week BAPS training program on postural stability and ankle proprioception in subjects with unilateral FAI. The hypothesis was that the 12-week BAPS training program would improve postural stability, as shown by a reduced mean radius of COP on unilateral standing, and improve ankle proprioception, as shown by better active and passive joint reposition senses.

2. Methods

2.1. Subjects

This study measured single-leg postural stability and ankle reposition senses for each limb in a group of young

subjects with unilateral FAI before and after a 12-week ankle BAPS training program. Twelve university students (4 females and 8 males) with unilateral FAI volunteered as subjects for this study and signed informed written consent before study participation, as approved by the human subjects ethics committee of our institution. Mean subject age was 20.08 ± 1.38 years, mean subject height was 172.33 ± 4.52 cm and mean subject body mass was 67.67 ± 4.89 kg. Limb dominance was determined by three functional tests: ball kick, step-up, and balance recovery (Lin et al., 2008) and 7 FAI occurred in the dominant limb and 5 FAI occurred in non-dominant limbs.

Before testing, all the subjects were not actively involved in any physical activity and were not in any exercise training or rehabilitation program. All subjects completed a questionnaire by self-report; for each subject, the non-injured limb (NI) had no history of ankle inversion sprain or lower extremity pathology, including fracture or arthritis. In addition, the passive and active ranges of ankle inversion, eversion, plantar-flexion, and dorsi-flexion were painless.

To be characterized as having unilateral FAI, subjects had to satisfy the following criteria: (1) experienced at least one significant lateral (inversion) ankle sprain of either the right or left ankle, but not both, in which the subject was unable to bear weight or was placed on crutches, within the last year; (2) no reported history of bone fracture to either ankle; (3) suffered at least one repeated injury or the experience of a feeling of instability or “giving way” in either the right or left ankle, but not both; (4) not undergoing any formal or informal rehabilitation of the unstable ankle; and (5) no evidence of mechanical instability, as assessed using an anterior drawer test (Kaminski et al., 2003; Caulfield and Garrett, 2004). All the inclusion/exclusion physical examinations and criteria tests in this study were performed by the same investigator, who was a physician with certified athletic trainer license in our country.

2.2. Postural stability assessment

Subjects stood barefoot in a comfortable upright posture with their eyes open or closed for a single-leg standing test on a force platform (Advanced Mechanical Technologies Inc, Waltham MA, USA). The force platform was linked via a 16-bit analog to digital converter (MP100, BIOPAC System Inc, Goleta CA, USA) to an IBM-compatible PC loaded with Acknowledge (version 3.8) for data collection and analysis. Subjects were instructed to lift the limb that was not being tested by flexing 90° at the hip and knee joints with both arms hanging relaxed at the sides. The subjects were asked to stand as motionless as possible, looking straight ahead at a point on the wall 65 cm away. Displacements of the COP were recorded at a frequency of 100 Hz and filtered with a 10 Hz low-pass cut-off frequency (DA100C, BIOPAC System Inc, Goleta CA, USA). The mean radius of the COP (the mean excursion of the COP in any direction), the average of all displacements of the

COP at each sampling interval from the mean position of the COP, was used as the balance parameter to describe the subjects' balance stability (Lee and Lin, 2007). Increased values in the mean radius of the COP suggest decreased postural control, whereas a decrease value is thought to present increased postural stability (Le Clair and Riach, 1996). Tests of single-leg standing balance have been demonstrated to have moderate to excellent group reliability, and therefore could be used to compare the balance performance of different groups of subjects (Birmingham, 2000).

After two practice trials with a 1-minute rest period before each testing, subjects were asked to perform the single-leg standing test on each limb for both "eyes open (EO)" and "eyes closed (EC)", alternately. Once the subjects were in this position and stated that they were ready, data collection was initiated. All the subjects performed three trials, 10 s in duration, with a 1-minute rest period between trials in order to avoid fatigue. The data from 3 trials of the mean radius of the COP during EO and EC testing for each limb (FAI and NI) were averaged to create representative index for the postural stability assessment of each subject. It has been reported that a testing duration of 10 seconds is reliable when assessing postural stability in a single-leg stance (Le Clair and Riach, 1996).

2.3. Ankle proprioception assessment

Active and passive reposition senses (ARS and PRS) were assessed using a Biodex 3 isokinetic dynamometer and the Biodex Advantage Software Package (Biodex Medical System, Inc, Shirley, NY, USA). Each subject was positioned supine on the associated chair with the lower limb parallel to the floor. This setup enabled proper placement of the subject's bare foot into the Biodex ankle inversion/eversion apparatus with the ankle plantar flexed 25°. After the subject's foot was properly aligned with the axis of the isokinetic dynamometer according to manufacturer's guidelines, a very small wrap was placed around the proximal tibiofibular joint and forefoot to provide stabilization and reduce cutaneous receptor input. Subjects were blindfolded during all the testing procedures to eliminate visual feedback. Three positions were tested: 15° of inversion, 0° subtalar neutral, and 10° of eversion from maximum inversion (Bernier and Perrin, 1998). Prior to being tested, each subject received a practice session followed by a 30-second rest period, and subsequently began the passive and active reposition test. The subjects were tested twice at each of the three test positions.

For passive reposition testing, the subject's foot was first passively moved by the dynamometer from each individual maximal inversion range-of-motion into the three testing positions at 2°/s. In the practice session, the dynamometer was stopped at each test position for 5 s, with each subject instructed to concentrate on the position of the foot. In the test session, the foot was passively moved just as it was in the practice session. The subject was instructed to use a

hand-held switch button to make a trigger signal when he or she felt the test position had been reached. In order to avoid the possible timing-prediction learning effects (e.g., at the angular velocity of 2°/s, it takes 5 s to move from 0° to 10° eversion), the subject does not know the actual velocity setting for the PRS proprioception test.

The active reposition test was performed in the same manner, except that after having the foot passively placed in the test position and moved to maximal eversion, the subject was asked to voluntarily move the foot actively back to each test position in sequence. The dynamometer produced an audible signal and was held for 5 s while the ankle was rotated to each pre-selected test position during the test. During the test session for the active ankle joint reposition test, the subject was asked to concentrate on their ankle and actively move their ankle to each test position in the same order as the practice session. In a similar manner to the passive reposition test, the subject was also requested to use a hand-held switch to make a trigger signal when he/she voluntarily moved his/her ankle to each of the pre-selected test positions within 60 s. The angular speed for the active reposition test was set at 500°/s (the highest speed setting for the proprioception test in Biodex III) to minimize additional resistance from the testing apparatus. Joint reposition sense assessments were recorded in absolute degrees of error between the pre-selected angle and the repositioned angle. Previous studies have demonstrated that there is no significant difference between the three test positions (Bernier and Perrin, 1998; Szczerba et al., 1995). Therefore, the absolute values of the three trials for each test position were averaged to create a representative index for the assessment of ARS and PRS of ankle proprioception for each limb.

2.4. BAPS training procedures

The training device used in this study was the biomechanical ankle platform system (BAPS; Spectrum Therapy Products, Jasper, MI, USA) because it is commonly used in athletic training and recommended for a clinical rehabilitative setting. The BAPS consists of a reversible platform, five hemispherical attachments, weight rods and weights and a storage rack. One of five differently sized half spheres can be manually attached to the surface of the platform to adjust the maximal tilt angle. The five levels challenge subjects to keep single-leg postural control while maintaining a steady stance or performing various range-of-motion exercises with their FAI limb.

Subjects began training on the BAPS at the lowest level (one out of five). Before the first training session, five rehabilitative exercises (anterior–posterior cycles, medial–lateral cycles, clockwise rotation, counterclockwise rotation, and single-leg stability, Table 1) were demonstrated to each subject (Riemann et al., 2003). All the training sessions and exercises were supervised by the investigators and conducted in the university's athletic training room. Each training sessions were approximately 20 min in length

Table 1
Exercises and the progression test for BAPS training

Category	Components	Setting	
Exercise	Anterior–posterior cycles	3 sets/10 rep	Move the platform back and forth in the sagittal plane in a controlled manner
	Medial–lateral cycles	3 sets/10 rep	Move the platform side to side in the frontal plane
	Clockwise rotation	3 sets/10 rot	Move the platform constant contact with the floor in a clockwise direction
	Counterclockwise rotation	3 sets/10 rot	Move the platform constant contact with the floor in a counterclockwise direction
	Single-leg stability	3 times/10 s	Try to keep the platform as level as possible
Progression	Board control	10 s	Move the platform in a clockwise/counterclockwise direction on command
	Board Level	7 s	Keep the platform as level as possible

rep: repetition; rot: rotation; s: second.

and the training period was 12 weeks long at a frequency of three times per week (36 sessions in total). Minimal use of upper body stabilization was encouraged by instructing the subjects to try to keep their hands on their iliac crests during all exercises.

This study only trained the FAI limb. Each subject's FAI limb was positioned on the platform according to the manufacture's recommendations. The first exercise was an anterior-to-posterior movement. Subjects were directed to move the board back and forth in the sagittal plane in a controlled manner, touching the floor with only the front and back aspects of the board. The subjects performed this exercise for three sets of ten repetitions, in which an anterior-to-posterior cycle counted as one repetition. The next exercise was similar, except that the board was positioned medial-to-lateral. Subjects were told to move the board side to side in the frontal plane, touching the floor with only the medial and lateral aspects of the board. The subjects performed this exercise for three sets of ten repetitions, in which a medial-to-lateral cycle counted as one repetition.

The next sets of exercise were clockwise and counterclockwise movements. Subjects were told to move the board in a clockwise direction for ten rotations, ensuring that it remained in constant contact with the floor as it was rotated. Similar instructions were given for the counterclockwise movement. The subjects alternated clockwise and counterclockwise movements until they had performed three sets of ten rotations in both directions. The final exercise, single-leg stability, required subjects to try to keep the board as level as possible for ten seconds while standing on one leg for three times. A 30-second rest period was allotted between exercises.

Similarly to previous studies (Hoffman and Payne, 1995; Riemann et al., 2003), subjects were individually advanced to the next level upon completing a progression test. Two progression tests were used; one tested the subject's ability to move the BAPS without losing postural control, and the other tested the subject's ability to keep the BAPS as level as possible. The objective of the first test was to move the board in a clockwise direction for ten seconds, and on command, to switch the board rotation direction to counterclockwise for 10 s without using external support, touching down with the non-testing limb or touching the testing limb with the non-test limb. Subjects who completed this

test were advanced to the next level for the anterior-to-posterior, medial-to-lateral, clockwise and counterclockwise exercises. The objective of the second test was to keep the board as level as possible for 7 s without using external support and without touching down with the non-test limb. If the subjects completed this progression test, they were advanced to the next level for the single-leg stability exercise.

2.5. Statistical analysis

Four 2×2 (limb $\times$ training) repeated measures analyses of variance (ANOVAs) were conducted to determine if any difference existed between limbs (FAI versus NI) and time of testing (pre-training versus post-training) for EO, EC, ARS and PRS. The mean radius of COP excursion during EO, the mean radius of COP excursion during EC, the absolute error from the reference angle during ARS, and the absolute error from the reference angle during PRS were dependent variables, and training was the independent variable. Statistical significance was set at the $P < .05$ level. All statistical analyses were conducted using SPSS for Microsoft Windows, Release 10.0 (SPSS Inc, Chicago, IL, USA).

3. Results

The final levels of BAPS training arrived at by the subjects all reached at least level 3 (of five levels), while eight subjects reached level 4 over the course of 12 weeks. The results of the ANOVA for mean radius of COP excursion during the EO condition (Fig. 1) indicated a main effect for training ($F = 18.87$, $P < .05$), and for limb ($F = 26.39$, $P < .05$), with post-training being better than pre-training, and with the FAI limb being more improved than the NI limb. In addition, There was an interaction between the variables of limb and training ($F = 11.44$, $P < .05$). Eta, an index of the effect size, for training, limb, and training $\times$ limb were about 0.55, 0.61, and 0.46, respectively, representing large effects.

The results of the ANOVA for mean radius of COP excursion during the EC condition (Fig. 2) indicated a main effect for training ($F = 30.77$, $P < .05$) and for limb ($F = 45.90$, $P < .05$), with post-training being better than pre-training, and with the FAI limb being more improved than the NI limb. In addition, there was an interaction

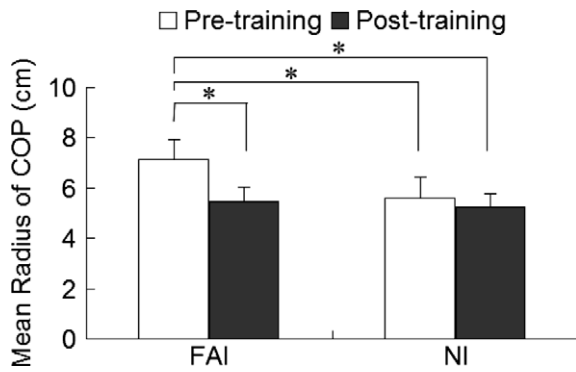


Fig. 1. Mean radius of COP excursion during the EO condition during the pre-training and post-training periods of the 12-week BAPS training program. * denotes a significant difference from the respective value, $P < 0.05$. FAI: functional ankle instability limb; NI: non-injured limb.

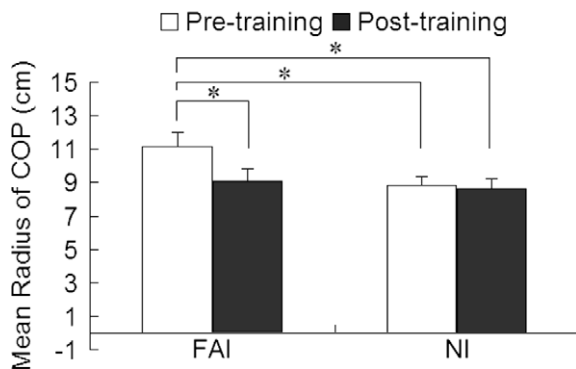


Fig. 2. Mean radius of COP excursion during the EC condition during the pre-training and post-training periods of the 12-week BAPS training program. * denotes a significant difference from the respective value, $P < 0.05$. FAI: functional ankle instability limb; NI: non-injured limb.

between the variables of limb and training ($F = 21.97$, $P < .05$). Eta for training, limb and training $\times$ limb were about 0.64, 0.71, and 0.57, respectively, representing large effects.

The results of the ANOVA for absolute errors from the pre-selected angle during the ARS test (Fig. 3) indicated a main effect for training ($F = 14.70$, $P < .05$) and for limb ($F = 19.87$, $P < .05$), with post-training being better than pre-training, and with the FAI limb being more improved than the NI limb. In addition, there was an interaction between the variables of limb and training ($F = 5.37$, $P < .05$). Eta for training, limb and training $\times$ limb were about 0.50, 0.56, and 0.33, respectively, representing large effects.

The results of the ANOVA for absolute errors from the pre-selected angle during the PRS test (Fig. 4) indicated a main effect for training ($F = 10.21$, $P < .05$), and for limb ($F = 9.31$, $P < .05$), with post-training being better than pre-training, and with the FAI limb being more improved than the NI limb. In addition, there was an interaction between the variables of limb and training ($F = 10.83$, $P < .05$). Eta for training, limb, and training $\times$ limb were about 0.43, 0.42, and 0.44, respectively, representing large effects.

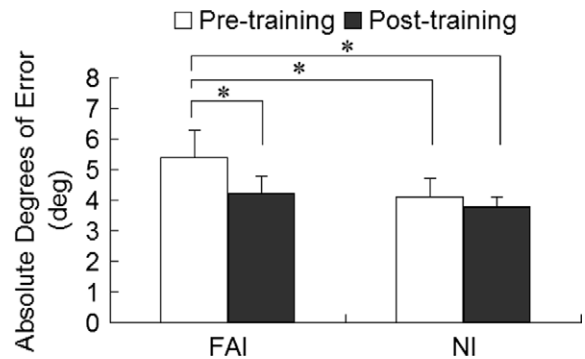


Fig. 3. Absolute degrees of error for ARS tests during the pre-training and post-training periods of the 12-week BAPS training program. * denotes a significant difference from the respective value, $P < 0.05$. FAI: functional ankle instability limb; NI: non-injured limb.

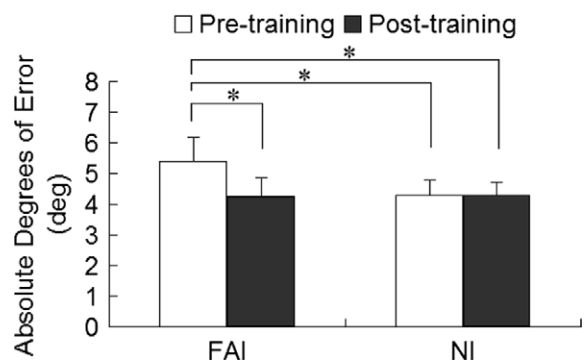


Fig. 4. Absolute degrees of error for PRS tests during the pre-training and post-training periods of the 12-week BAPS training program. * denotes a significant difference from the respective value, $P < 0.05$. FAI: functional ankle instability limb; NI: non-injured limb.

4. Discussion

Previous follow-up studies have demonstrated the residual after the ankle inversion trauma complaints might last 6–18 months (Konradson et al., 2002; Lentell et al., 1995). The use of a 12-week BAPS training program in this study was based on suggestions from a previous study which demonstrated that wobble board training for a 12-week period after partial sprains was effective in reducing the number of recurrent distortions and minimizing posttraumatic edema (Wester et al., 1996). Moreover, a later study also reported that ankle injury resulted in reduced ankle strength and postural control at 6 weeks, but that these variables had normalized at 4 months, independent of the rehabilitation (Holme et al., 1999). The results of the current study demonstrate that the mean radius of COP on unilateral standing (postural stability) and the absolute error from a pre-selected ankle angle (ankle proprioception) in the FAI limb were significantly reduced after a 12-week BAPS training program.

To avoid the possible ceiling effect of the BAPS training, this study used a training paradigm with progressive task difficulties as the subject's performance improved in the progression test (Hoffman and Payne, 1995; Riemann

et al., 2003). This differs from previous studies that reported either no progression of the training exercise (Clark and Burden, 2005; Wester et al., 1996) or that based exercise progression on standardized time frames (Bernier and Perrin, 1998; Paterno et al., 2004). Although the training design for non-progressive programs and progressive programs is different, the results seemed to agree well because similar tasks and exercises were also utilized in both programs, and was unable to dissect out which design provided better results. The decision in this study to use a progressive test in this study was based on the clinical-practice technique of adjusting rehabilitation programs according to the unique characteristics of individuals. Thus, although a participant may reach a plateau in test-performance progress, it may still be beneficial to increase the difficulty of the exercises after a particular period of time.

Progressive testing in this study involved increasing the size of the hemisphere on the underside of the multi-axial platform. A larger hemisphere increases the maximal angle at which the board can be tilted, which in turn influences the quantity of lower leg muscle activation (Riemann et al., 2003). Previous studies with a non-progressive training design have demonstrated positive effects on postural stability (Paterno et al., 2004) and muscle onset latency (Clark and Burden, 2005), but a lack of significant improvement in joint position sense (Bernier and Perrin, 1998). The results of this study were similar to the findings of previous studies on postural stability, but further demonstrated that the joint position sense can be improved in the FAI limb by progressive BAPS training.

Moreover, the pathogenesis of FAI is complex, and loss of proprioception, which results in a lack of balance and joint position sense, is considered to be particularly important (Clark and Burden, 2005). A progressive training program should be more suitable for the different magnitudes of the deficit among individuals with FAI limb. In addition, an exercise program with a progressive test might be more challenging, more fun, and might keep participants motivated during the training sessions; it might also improve the adherence to the program more than a non-progressive program.

The improvements in postural stability and ankle proprioception in the FAI limb after 12 weeks of BAPS training in this study are consistent with most studies involving a wobble board and/or balancing training programs (Paterno et al., 2004; Hoffman and Payne, 1995; Rozzi et al., 1999), but are in contrast to some studies (Riemann et al., 2003; Verhagen et al., 2005; Bernier and Perrin, 1998). Although conflicting reports have been published about whether postural control is disrupted in ankles with functional instability ankle (Hertel, 2000; Riemann, 2002), the reason for the differing findings might be the variation in the magnitude of the deficits among individuals. For example, one individual might have large balance deficits, while another individual may have more proprioception deficits (Verhagen et al., 2005). This suggests that the training program utilized in this study effectively stimulated the

centrally mediated neuromuscular control mechanisms responsible for preserving balance and posture.

Postural stability has been suggested to be mediated by the same peripheral afferent mechanism that mediates joint proprioception, but it may be more representative of lower extremity function when compared to assessments performed in a nonweight-bearing position. BAPS training is a form of closed-chain exercise which might induce maximal stimulation of joint, muscle and plantar-surface mechanoreceptors found about the lower limb. During BAPS training, participants in this study had to consciously manipulate their ankle position and keep balance to coordinate the various exercises. A previous study demonstrated that a 12-week wobble board training program after partial sprains was effective in reducing the number of recurrent distortions and minimizing the posttraumatic edema (Wester et al., 1996). This study showed that the ankle joint reposition sense was strengthened after a 12-week BAPS training program. A literature search revealed that only one previous study has examined the effects of a 4-week BAPS training program on conscious appreciation of kinesthesia, but that study did not demonstrate significant improvements (Riemann et al., 2003). Another study (Bernier and Perrin, 1998), examining the effects of 6 weeks of coordination training on conscious perception of proprioception and active and passive joint position sense also failed to reveal any improvements. Potential explanations for these discrepancies might be the subjects used (injured or healthy), the setting of the training programs (duration and frequency) or the assessment techniques.

The most noteworthy result from this study was that the 12-week BAPS training program produced significant improvement in ankle proprioception. In comparison to previous studies that trained healthy (Hoffman and Payne, 1995; Riemann et al., 2003) or injured subjects (Wester et al., 1996; Clark and Burden, 2005; Bernier and Perrin, 1998) on multi-axial surfaces for different training period of weeks, e.g., 4 (Clark and Burden, 2005; Riemann et al., 2003), 6 (Bernier and Perrin, 1998; Paterno et al., 2004; Verhagen et al., 2005) or 12 weeks (Wester et al., 1996), where significant improvements in postural stability (Bernier and Perrin, 1998; Paterno et al., 2004) or reduced residual symptoms following ankle injuries (Wester et al., 1996) were reported, the results of this study demonstrated that the setting and the specific training program were effective in improving the sensorimotor functions of the FAI and could be used for rehabilitation. Although the measured proprioception improvements for ARS and PRS were only decreased by 1.2° and 1.1° (the absolute error decreased from 5.4° before training to 4.2° after training), respectively, they might still be of physiological relevance and clinical importance, because a previous study reported that an inversion error greater than 7° would equal a 5 mm drop of the lateral border of the foot, which would create foot contact, hyper-inversion and the potential for injury (Konradsen and Magnusson, 2000).

The enhanced ankle proprioception demonstrated in this study may be due to the improved mechanoreceptor function, which could lead to restoration of neuromuscular control of the joint. This restoration of the neuromuscular feedback loop may be a key factor that causes a positive outcome, i.e., a return to more stable or pre-injury levels (Lentell et al., 1995; Rozzi et al., 1999). In addition, coordination training also promotes somatosensory afferent inflow to the central nervous system, strengthening, retraining or possibly creating compensatory afferent pathways (Clark and Burden, 2005; Lephart et al., 1997; Paterno et al., 2004; Riemann et al., 2003). In addition, balance and/or coordination training on a multi-axial support surface require specific muscle-stabilization mechanisms and faster stabilization that originate from ankle mechanoreceptors (Riemann et al., 2003). Since BAPS training is performed in the weight-bearing position, it may be considered a suitable method for reestablishing neuromuscular control, and therefore for improving functional stability of the ankle joint.

This study has several possible limitations that need to be considered. First, even with relative large η values, there was a relatively small sample size in this study ($n = 12$), which could lead to a type I statistical error, meaning that the findings cannot be generalised to the entire population. Secondly, even with strict inclusion criteria and a narrow age range (20.08 ± 1.38 years), subjects in this study included both male and female students and still presented with a variety of potentially uncontrollable confounding factors, such as degree of injury severity, level of physical activity, and gender differences in lower extremity kinematics. Thirdly, although the subjects did not know the actual timing setting for the PRS proprioception test and similar error values were also founded during the ARS and PRS tests. However, there might still be a possibility that subjects have learned the timing from the practice trials. These factors would potentially prohibit precise conclusions from being drawn concerning adaptations.

Although it has been shown that ankle proprioception can be improved after 12 weeks of BAPS training in the FAI limb, there is still some controversy about the period of the training program needed for positive adaptation to occur and for clinical practice. In addition, it is difficult to draw conclusions about the long-term effects of this training without a follow-up test. Therefore, further research is needed to investigate the point at which BAPS training-induced improvements in postural stability and ankle proprioception begin to plateau. Moreover, most rehabilitation training programs and/or exercise intervention studies have only examined the improvements before and after training; therefore, a study design with a follow-up test is strongly needed to demonstrate the sustained effects of these specific training program, and may be more clinically relevant. Furthermore, due to the subjects in this study were not actively involved in any physical activity or exercise training, therefore, further studies are still needed to evaluate the positive effects of this training intervention

in the athletes population in terms of routine practice and/or during rehabilitation period.

5. Conclusions

The findings of this study support the use of a 12-week BAPS training program as part of the rehabilitation for subjects with unilateral FAI. Results show that this training improved postural stability and ankle proprioception in the FAI limb. These improvements in postural stability reflect improved neuromuscular ability and better functional joint stability, because ankle proprioception showed the same positive training effects after BAPS training.

Acknowledgements

The authors would like to express their best honors to anonymous reviewers for their valuable comments and suggestions to improve the quality of earlier version of this manuscript. This study was supported by grants from National Science Committee (NSC 95-2518-C-134-008-H & 96-2413-H-134-008) and National HsinChu University of Education, Taiwan, ROC.

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